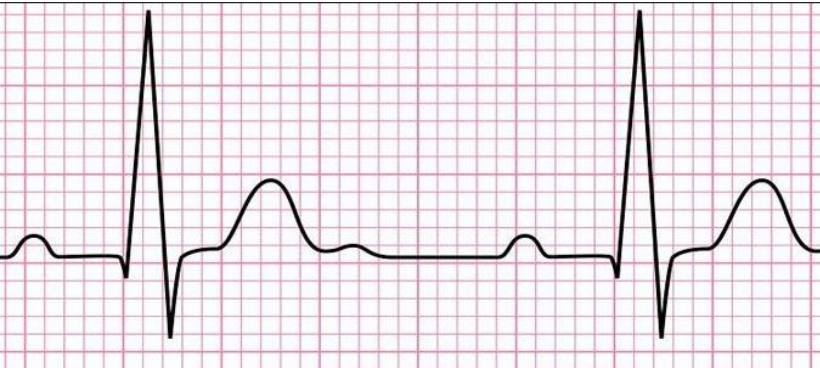
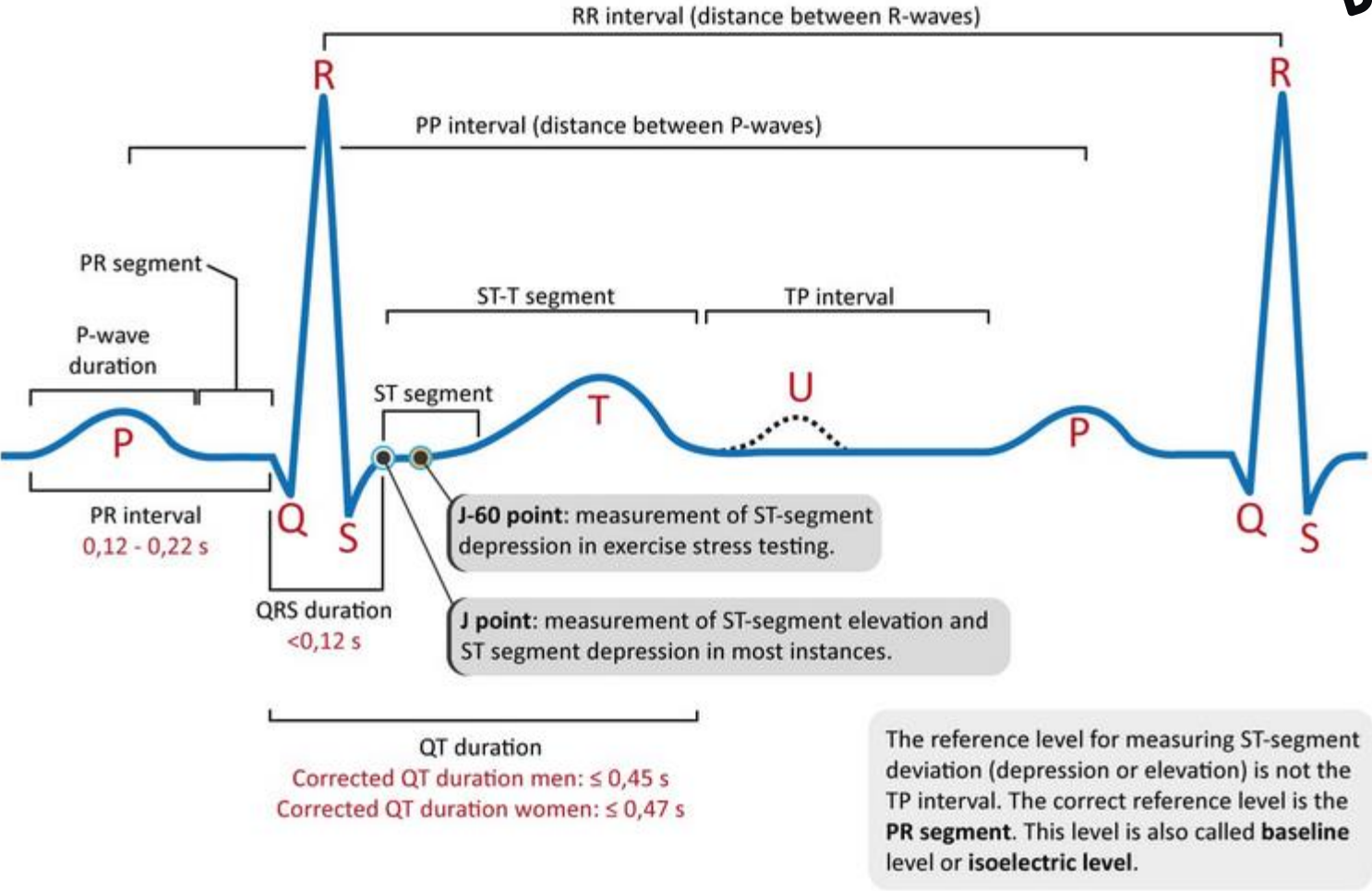


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ECG Waveform Measurements

Standard ECG Waveform

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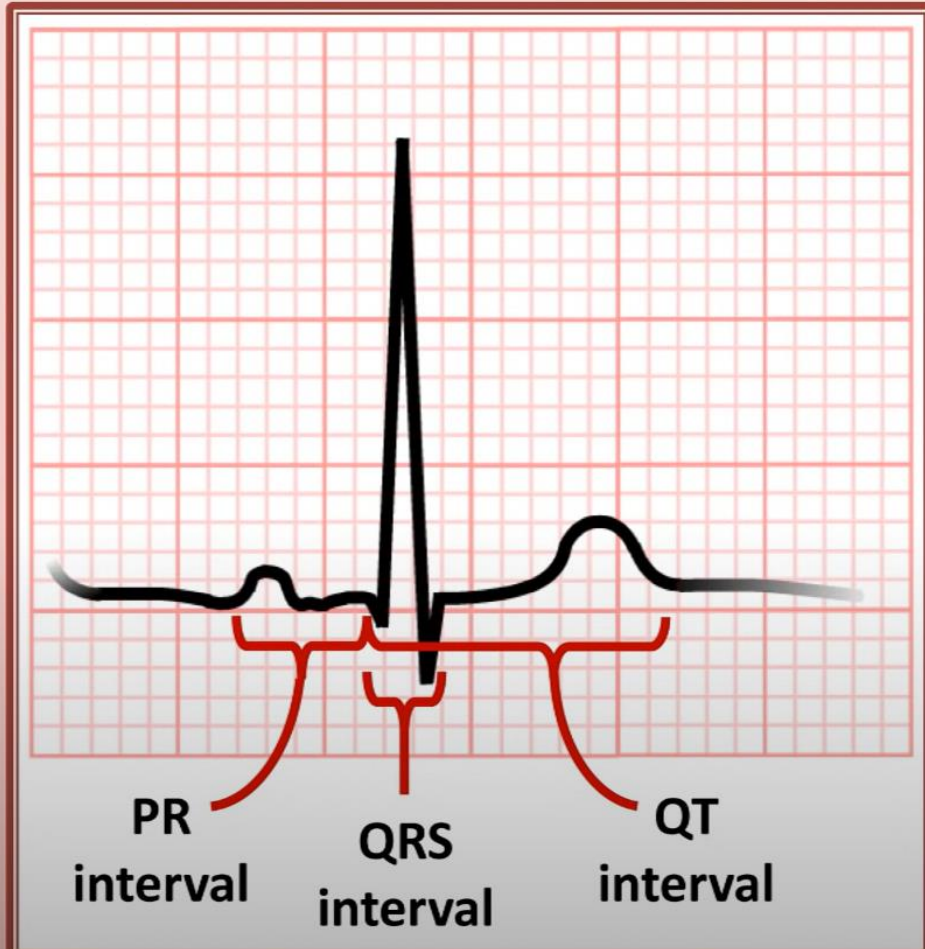


The reference level for measuring ST-segment deviation (depression or elevation) is not the TP interval. The correct reference level is the **PR segment**. This level is also called **baseline level** or **isoelectric level**.

ECG Waveform-Interval Significance

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Intervals

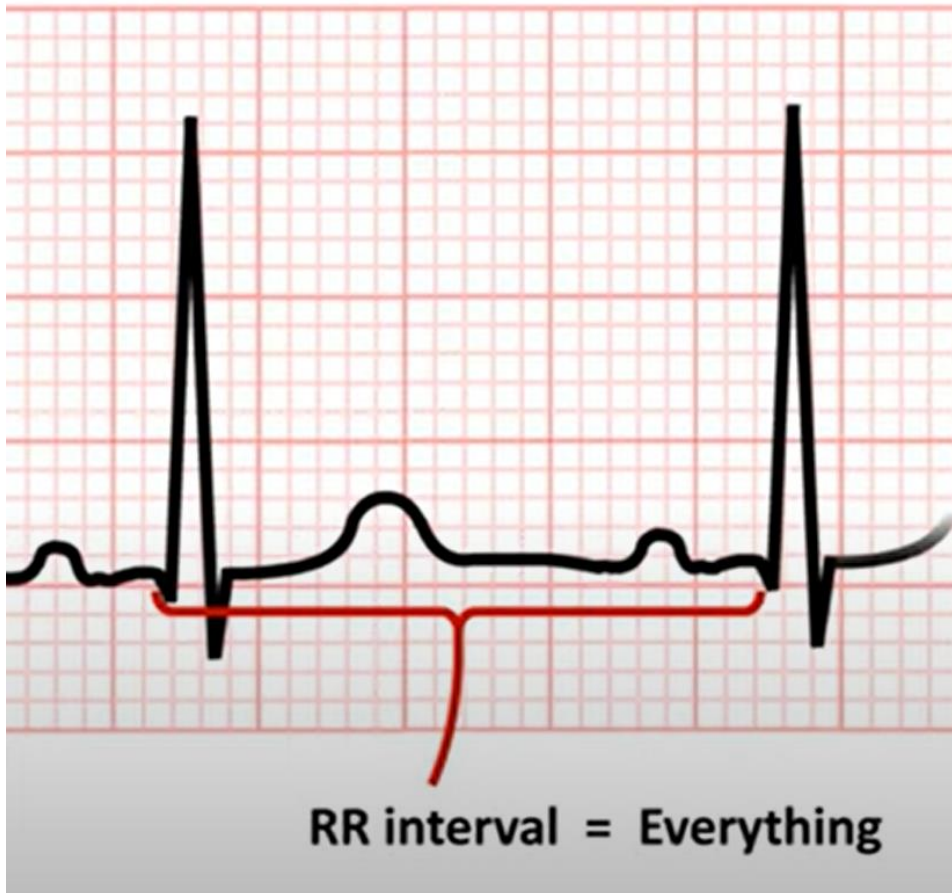


Interval	Normal Duration	Significance of abnormal duration
PR	120-200ms	<120ms: Abnormal connection between atria and ventricles >200ms: Delayed conduction through AV node
QRS	$\leq 120\text{ms}$	>120ms: Abnormal sequence of ventricular depolarization, or excessive ventricular mass
QT	$\leq \frac{1}{2}$ RR interval (an approximation)	> $\frac{1}{2}$ RR interval: Delayed ventricular repolarization

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ECG Simple Heart Rate Measurement

ECG Waveform-Simple Heart Rate Measurement



The RR interval is the duration of all waveforms and segments in a cardiac cycle.

Normal resting heart rate for a healthy adult is 60-100 Beats Per Minute (BPM)

One simple method to measure heart rate using an ECG signal is to measure the duration between two identical points of consecutive ECG waveforms such as the R-R interval. Take this duration and divide it into 60. The resulting equation would be:

Rate = $60 / (\text{R-R interval})$.

e.g., RR interval = 0.8 seconds. Heart rate = $60 / 0.8 = 75$ BPM

However, for our ECG, it would probably be more accurate to take this measurement over multiple RR intervals.

A simpler, not quite as accurate way may be to count the number of R-wave peaks over a given amount of time. Keep in mind that the amplitude of a healthy person's R-wave can vary significantly.

e.g., 75 R-wave peaks over 60 seconds is roughly 75 BPM. 33 R-wave peaks in 30 seconds is roughly 66 BPM, etc.

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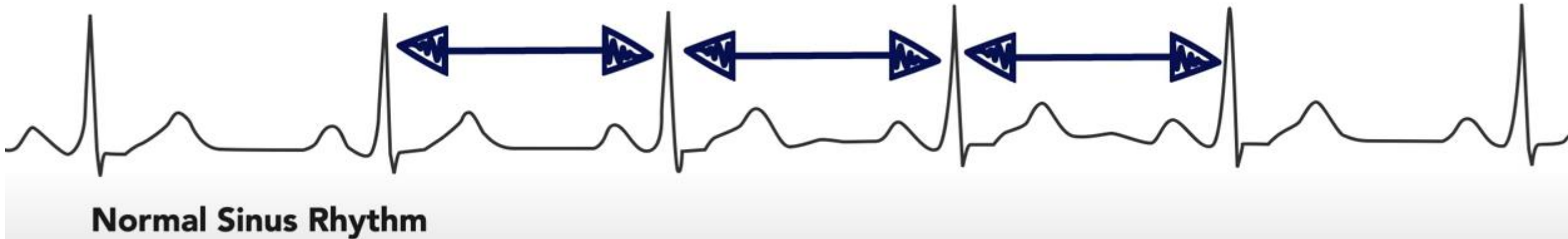
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Atrial Fibrillation (Afib) Heart Rhythm

ECG Waveform-Normal Heart Rhythm

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EQUAL INTERVALS

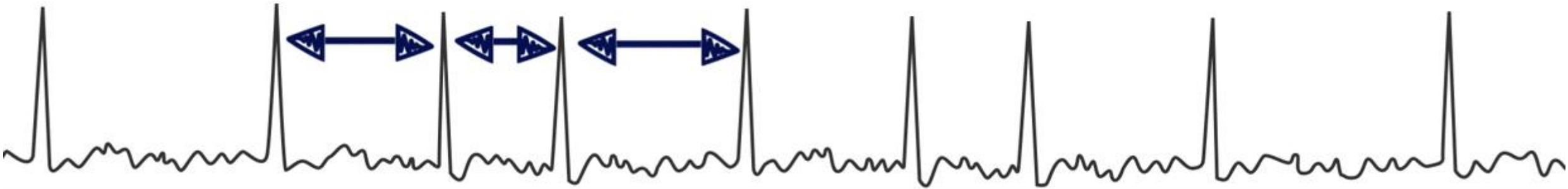


One characteristic of a normal rhythm is having equal RR intervals

ECG Waveform-Atrial Fibrillation (Afib) Heart Rhythm

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Irregular Intervals



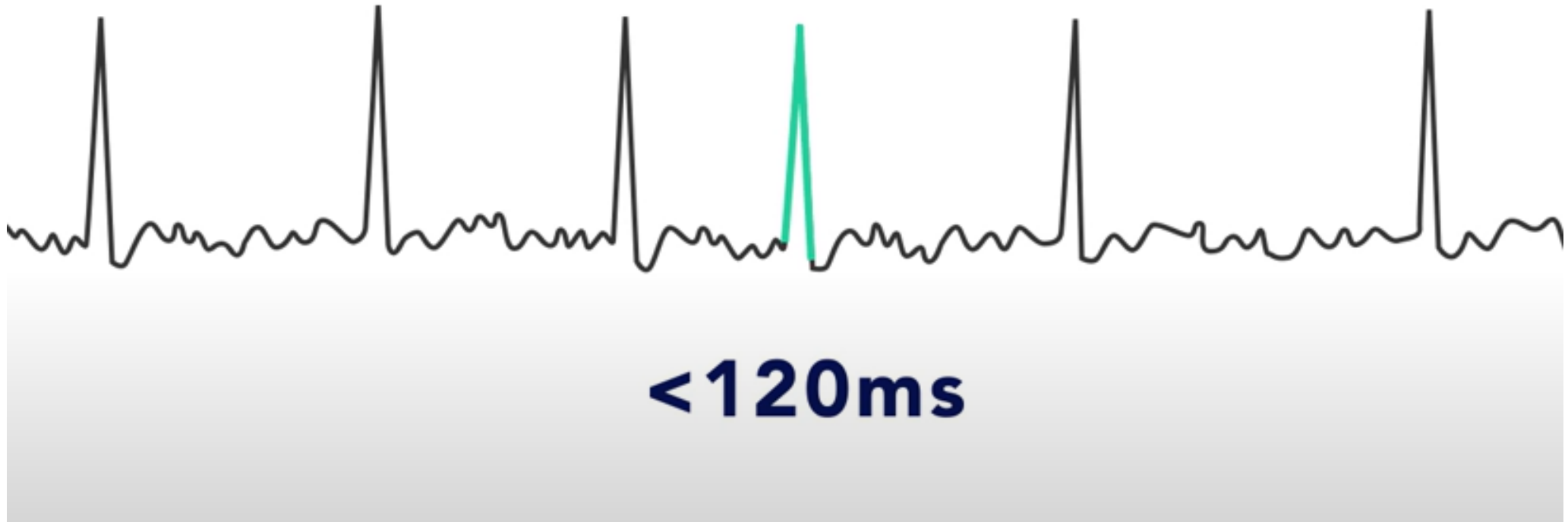
Ventricular beats 90 - 170 bpm

This is an irregular RR interval. The common indication of atrial fibrillation is irregularity of R peak-to-R-peak time interval, or RR interval

ECG Waveform-Atrial Fibrillation (Afib) Heart Rhythm

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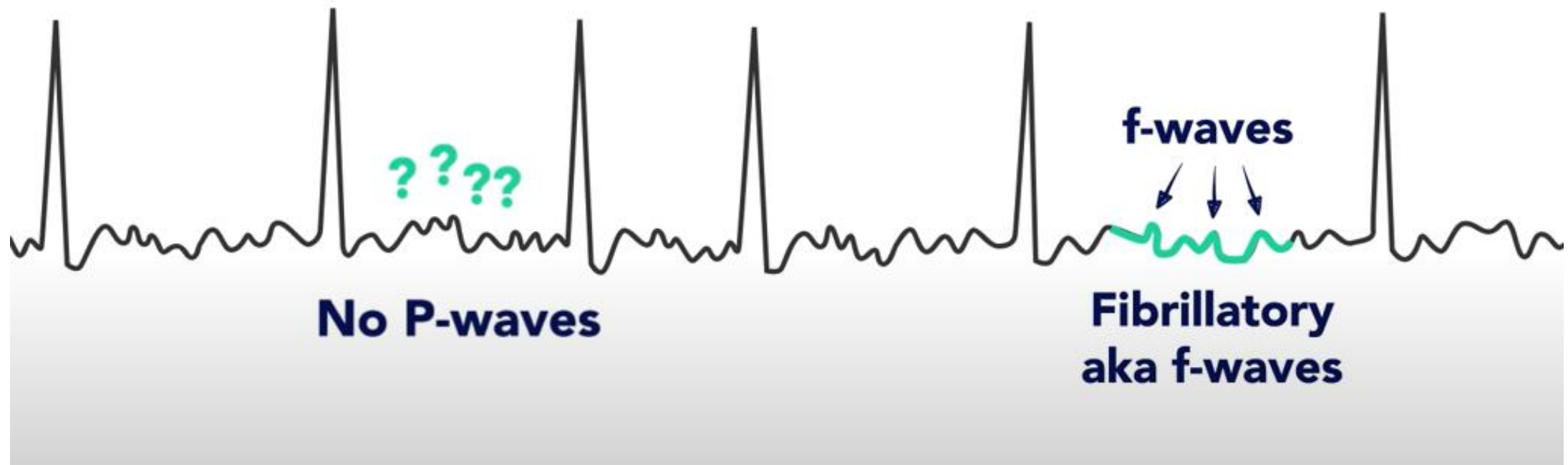
Narrow QRS



Other characteristics of Afib-Note: Narrow QRS by itself may indicate a normal heart rhythm. So, by itself, this does not necessarily indicate an Afib condition.

ECG Waveform-Atrial Fibrillation (Afib) Heart Rhythm

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Other characteristics of Afib

ECG Waveform-Atrial Fibrillation (Afib) Heart Rhythm

Afib Summary

- Irregularly irregular rhythm

and

- No distinct P-waves

and

- Presence of f-waves

and

- Narrow QRS complexes (less than 120 mS interval and with above issues)

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ECG Waveform-Sinus Bradycardia

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ECG Reference Design should only inform the user of a lower than normal heart rate and not try to diagnose this condition

- Heart rate of less than 60 bpm when patient is awake and active (possible exception-runners and athletes)
- Heart rate is less than 40 bpm
- P Waves precede each QRS complex
- PR interval is normal
- QRS complex is normal
- Conduction is normal
- Rhythm is regular

ECG Waveform-Sinus Tachycardia

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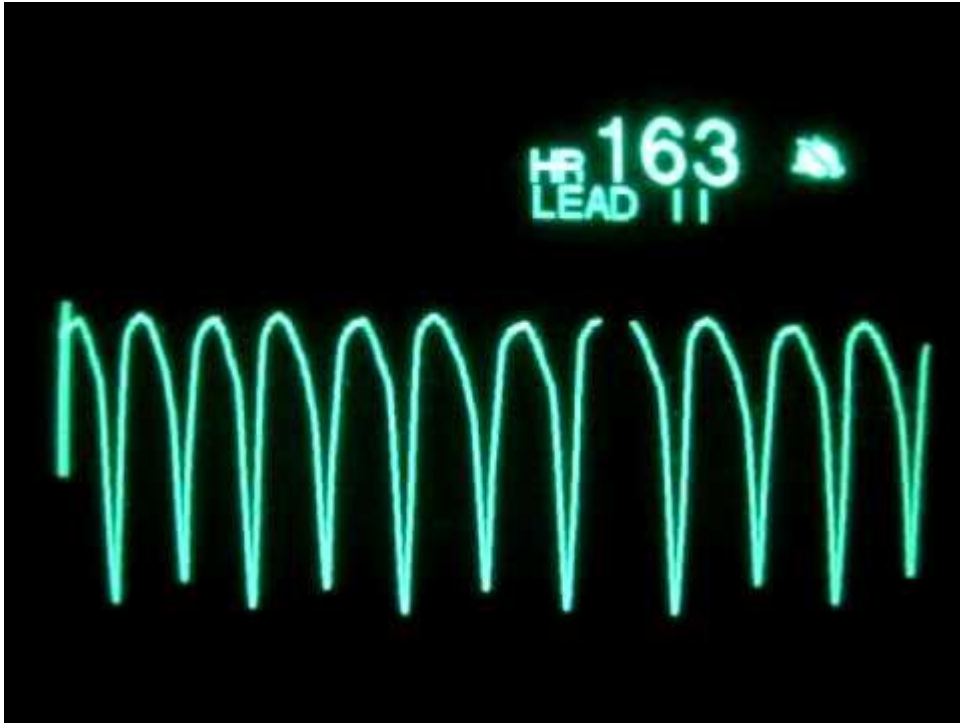


- Heart rate is 100 -180 bpm from the sinus node
- P Waves precede each QRS complex
- PR interval is normal
- QRS complex is normal
- Conduction is normal
- Rhythm is regular

ECG Reference Design should only inform the user of a higher than normal heart rate and not try to diagnose this condition

ECG Waveform-Ventricular Tachycardia

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ECG Reference Design should only inform the user of a higher than normal heart rate and not try to diagnose this condition

- Considered a medical emergency
- Heart rate is 100 -250 bpm
- P wave is blurred in the QRS complex, but the QRS complex has no associate with P wave
- PR Interval is not present
- QRS complex is wide and bizarre; T wave is in the opposite direction
- Rhythm is usually regular
- May start and stop suddenly